

Dunman High School
2021 Preliminary Examination Candidates' Check List- 9744

There will be a pause in examination after 1 hour 15 minutes for changeover of microscope. Please follow the instructions given by the Invigilator.

All candidates will have access to the materials for Question 1 throughout this examination.

Candidates with microscope on your bench **MUST** start with Question 2. You will **NOT** have access to the microscope after 1 hour 15 minutes. You may move on to Question 1 once you have finished working with the microscope. The other candidates will start with Question 1. After 1 hour 15 minutes, you will have access to the microscope and should move on to Question 2 as soon as possible.

All substances in the examination should be treated with caution. Suitable eye protection should be used where necessary.

Some chemicals/substances are placed in plastic vials/containers instead of beakers as stated in the question paper.

For Question 1

Each candidate will require

1	E
2	S
3	one beaker/container labelled hot water for collecting hot water
4	one beaker/container labelled tap water for collecting tap water
5	one beaker/container labelled ice water If the beaker/container does not contain ice and water, candidates may request for these or collect from a central dispensing point. All candidates can request or collect additional ice if required.
6	two 5 cm ³ syringes
7	five 250 cm ³ beakers for setting up water-baths
8	one glass rod (also used in Question 2)
9	five test-tubes
10	one test-tube rack to hold five test-tubes
11	one test-tube holder
12	one Bunsen burner, lighter, wire gauze and tripod
13	one thermometer
14	one beaker/container labelled waste (also used in Question 2)
15	one permanent marker (also used in Question 2)
16	one stopwatch
17	suitable eye protection and disposable gloves (also used in Question 2)
18	paper towels
19	30 cm ruler (also used in Question 2)
20	one clean dry cloth (e.g., for cleaning) (also used in Question 2)

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For Question 2

Each candidate will have sole, uninterrupted use of a microscope for 1 hour 15 minutes only.
For each candidate:

- the microscope must be set on the low-power objective lens
- the slide must **not** be left on the stage of the microscope.

Each candidate will require

1	one prepared slide, labelled MI (for students starting with Question 2)
2	one microscope (for students starting with Question 2)
3	two 5 cm ³ syringes
4	B
5	P
6	Q
7	R
8	three test-tubes
9	one test-tube rack to hold three test-tubes (also used in Question 1)
10	one glass rod (also used in Question 1)
11	one beaker/container labelled waste (also used in Question 1)
12	30 cm ruler (also used in Question 1)
13	one permanent marker (also used in Question 1)
14	suitable eye protection and disposable gloves (also used in Question 1)
15	paper towels
16	one clean dry cloth (e.g., for cleaning) (also used in Question 1)

Question 1

Preparation list

	Apparatus / Reagents / Chemicals	Quantity per candidate
1	[HH] Plant extract solution, in a 100 cm ³ glass beaker labelled E and GHS pictogram (see Note 1).	40cm ³
2	1% hydrogen peroxide solution in a 100 cm ³ glass beaker or plastic vial (B155), labelled S (see Note 2).	20cm ³
3	Access to tap water and sink	
4	Beaker or container (e.g., glass or plastic), suitable for collecting at least 250 cm ³ of tap water, labelled tap water	1
5	Beaker or container (e.g., glass or plastic), suitable for collecting at least 400 cm ³ of hot water, labelled hot water . The beaker should have a mark at a volume of approximately 400 cm ³ , labelled to show this volume. This mark and label should be added with a permanent marker if not already present. Candidates will collect hot water in this beaker during Question 1. Hot water should be available at a temperature of 50°C to 55°C, e.g. from thermostatically controlled water-baths. Candidates may need to collect additional hot water as required.	1
6	Approximately 400 cm ³ of a mixture of ice and water at a temperature of no more than 10 °C, in a beaker or container (e.g., glass or plastic) labelled ice water . This mixture should be provided to candidates just before they start Question 1 (see Note 3). Additional ice should be provided upon request.	
7	250 cm ³ glass beakers	5
8	5 cm ³ syringes	2
9	Test-tubes, to hold no more than 25 cm ³ volume, suitable for heating	5
10	Rack to hold 5 test-tubes and test-tube holder	1 each
11	Clean dry cloth (also used in Question 2)	1
12	Glass rod (also used in Question 2)	1
13	Bunsen burner, lighter, wire gauze and tripod	1
14	Thermometer, -10 °C to +110 °C	1
15	250 cm ³ empty glass beaker, or container (e.g., plastic), labelled waste (also used in Question 2)	1
16	30 cm ruler (cm/mm), clear plastic (also used in Question 2)	1
17	Permanent marker suitable for writing on glass (also used in Question 2)	1
18	Stopwatch	
19	Suitable eye protection and disposable gloves (also used in Question 2)	
20	Paper towels	

1 pair each

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Note 1 **[HH]** Plant extract solution in a glass beaker labelled **E**

Mung beans for the plant extract solution **E** must be soaked in tap water for 24 hours before the examination. Sufficient mung beans should be germinated to allow preparation of additional extracts if the activity of the extract is too low.

You must test the plant extract solution **E** in advance of the examination to allow time for preparation of a fresh extract, if necessary.

Plant extract solution **E** and hydrogen peroxide solution **S** must be prepared on the day of the examination and should be at room temperature at the start of the examination.

Instructions for preparation:

Dried mung beans have been provided for Centres' use. These mung beans must first be soaked for approximately 24 hours in tap water.

On the day of the examination, the plant extract solution can be prepared using 10g of soaked mung beans as follows:

- Put approximately 10 g of soaked mung beans into a blender with 100 cm³ of water. Liquidise until the mung beans have been broken into small pieces.
- Strain the mixture through a fine sieve or muslin and retain the solution. This may still contain small pieces of plant material. This is acceptable.
- This should produce at least 90 cm³ of plant extract solution **E**. This should be sufficient for two candidates.

To test the plant extract solution E:

- Put 1 cm³ of **E** into a test-tube.
- Put 1 cm³ of **S** into the same test-tube.
- You should begin to observe bubbles within 30 seconds. If the bubbles take longer than 30 seconds to appear, increase the mass of mung beans used to prepare the extract.

Note 2 1% hydrogen peroxide solution, in a beaker labelled **S**

6% hydrogen peroxide has been provided for Centres' use. To prepare 1% hydrogen peroxide solution, mix 20 cm³ of 6% (20 vol) hydrogen peroxide solution **[MH]** with 100 cm³ of distilled water. This gives a final volume of 120 cm³, which should be sufficient for at least five candidates.

Note 3 The ice mixture should be provided to candidates beginning with Question 1 at the start of the examination. Candidates beginning with Question 2 could receive the ice during the pause in examination (after 1 hour and 15 min) immediately after the microscopes are cleaned and transferred, or anytime they are ready to start on Question 1.

Question 2

Preparation list

Each candidate must have sole uninterrupted use of a microscope and slide **MI** for 1 hour 15 minutes only.

For each candidate:

- the microscope must be set up on the low-power objective lens
- slide **MI** must **not** be left on the stage of the microscope.

	Apparatus / Reagents / Chemicals	Quantity per candidate
		1 between 2
1	Microscope with an eyepiece graticule, as specified on page 2	1 between 2
2	Prepared slide, labelled MI (provided by Cambridge International). See Note 4.	10 cm ³
3	Distilled water, in a plastic vial (8 43) sealed with a lid labelled P	10 cm ³
4	0.1 mol dm ⁻³ sodium hydroxide solution, in a plastic vial (8 43) sealed with a lid labelled Q . See Note 5.	10 cm ³
5	0.1 mol dm ⁻³ hydrochloric acid, in a plastic vial (8 43) sealed with a lid labelled R . See Note 5.	20 cm ³
6	Hydrogencarbonate (bicarbonate) indicator, in a 100 cm ³ glass beaker or plastic vial (8 155), labelled B . See Note 5.	
7	5 cm ³ syringe	2
	8 Test-tubes to hold no more than 25 cm ³ volume	3
9	Rack to hold 3 test-tubes (also used in Question 1)	1
10	Glass rod (also used in Question 1)	1
11	250 cm ³ empty glass beaker, or container (e.g., plastic), labelled waste (also used in Question 1)	1
12	Access to tap water and sink	
13	30 cm ruler (cm /mm), clear plastic (also used in Question 1)	1
14	Permanent marker suitable for writing on glass (also used in Question 1)	1
16	Paper towels	8
17	Clean dry cloth (also used in Question 1)	1

Note 4 Prepared slide, labelled **MI** (supplied by Cambridge International)

On receipt of the slides, please check that they are labelled **MI** and that none of the slides are broken.

The material is **confidential** (so must **not** be disclosed to candidates) and the slides should **not** be viewed in advance of the examination.

Note 5 Commercial preparations of the chemicals listed in items 4 to 6 are provided for Centres' use.